

Ammar Yasser Abdallah

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PROFILE

As a passionate AI Engineer, I focus on building intelligent systems by applying machine learning NLP and optimization algorithms to solve real world problems. my experience spans search algorithms deep learning computer vision and modern AI techniques such as Transformers and RAG based systems I enjoy transforming complex ideas into practical AI solutions, with a strong emphasis on performance, scalability and continuous learning

EDUCATION

Computers and Data Science (AI) 2022 – present
Alexandria, Egypt

Yathreb Private School Alexandria, Egypt

LANGUAGES

Arabic ● ● ● ● ● English

SKILLS

Problem-Solving, Team Collaboration, Communication, Time Management, Critical Thinking, Leader Ship, Organizer, Python, Java, C++, HTML, SQL, next js, mongoDB, javaScript, TypeScript, nodeJS, nextAuth, Docker, Excel, Microsoft Office, Adobe premiere

COURSES

Machine Learning

Cloud Computing

Databases

Operations Research

Pattern Recognition

NLP

Embedded System

IOT

Databases

PROJECTS

Connect 4

AI-powered Connect-4 game using **Java and JavaFX**, integrating advanced search algorithms such as **Minimax** with **Alpha-Beta pruning** to enable intelligent gameplay and efficient decision-making. The project features an interactive graphical interface, multiple difficulty levels, and a custom heuristic evaluation function that allows the AI to predict and counter player moves effectively. This project strengthened my understanding of **adversarial search, game theory, and heuristic optimization**, while also enhancing my **Java programming** and **UI design** skills.

8-Puzzle Game

Developed an **8-Puzzle Game solver** in **Java** implementing classical **Artificial Intelligence search algorithms** such as **A***, **Uniform Cost Search**, and **Greedy Best-First Search** to efficiently find the optimal sequence of moves. The system uses heuristic evaluation functions like **Manhattan Distance** and **Misplaced Tiles** to guide the search process. A simple **graphical user interface** was designed to visualize puzzle states and solution steps in real time. This project enhanced my knowledge of **state-space search, heuristic design, and problem-solving optimization techniques** in AI, while improving my **algorithmic thinking** and **Java programming** skills.

Phone Book

Phone Book application in **Java** that allows users to **store, search, update, and delete contact information** through an interactive interface. Implemented **data structures** such as **HashMaps** and **ArrayLists** to efficiently manage and retrieve contact records. Added features like **alphabetical sorting, real-time search, and data validation** for enhanced usability and accuracy. This project improved my understanding of **object-oriented programming (OOP)** principles, **data management**, and **user interface design**, while strengthening my practical skills in **Java development**.

Dockerized Web Application with Database

Developed and deployed a **full-stack web application** integrated with a **relational database**, fully **containerized using Docker** for consistent and scalable deployment across environments. Designed the backend with **RESTful APIs** to handle CRUD operations and connected it to a **MySQL** database for reliable data storage and retrieval.

Automated Railway Station

Developed an **Automated Railway Station management system** designed to streamline **train scheduling, ticket booking, and passenger information handling**. The system automates core operations such as **arrival and departure tracking, seat availability management, and real-time updates** using efficient data structures and algorithms. Built with a focus on **modularity and scalability**, the project integrates both **frontend and backend components** to simulate real-world railway operations. This project enhanced my understanding of **system automation, database management, and software architecture**, while strengthening my skills in **problem-solving, programming logic, and process optimization**.

Traveling Salesman Problem (Genetic Algorithm)

Ant Colony Optimization (ACO) algorithm to solve **combinatorial optimization problems** such as the **Travelling Salesman Problem (TSP)**. The project simulates the behavior of real ant colonies to find optimal paths through **pheromone-based communication and probabilistic decision-making**. Designed the algorithm from scratch with adjustable parameters for **pheromone evaporation, intensity, and heuristic influence**, enabling performance tuning and result visualization.

Library With Database

Developed a **Library Management System** integrated with a **relational database** to efficiently manage **books, users, and borrowing records**. Implemented core functionalities such as **book issuance, return tracking, fine calculation, and search operations** using **SQL queries** and **CRUD operations**. The backend was connected to a **MySQL database** to ensure data consistency and integrity, while the interface was designed for ease of use and scalability.

Smart Home (IOT)

Developed a collection of **IoT-based smart system projects** that integrate **hardware automation, a mobile application, and a web interface** for seamless control and monitoring. Built systems such as **smart home automation** and **sensor-based monitoring** using **Arduino, ESP32, and Raspberry Pi** microcontrollers, connected via **Wi-Fi** and **MQTT protocols**. Designed a **mobile app** (using **Flutter / Android Studio**) and a **responsive website dashboard** (using **HTML, CSS, JavaScript, and REST APIs**) to provide real-time data visualization and remote device management. This project enhanced my skills in **embedded systems, IoT communication, full-stack development, and cross-platform integration**, demonstrating end-to-end implementation from hardware to cloud to user interface.

machine translation (English-Spanish)

Developed a **Machine Translation system** as part of an NLP project, implementing and comparing approaches based on **Recurrent Neural Networks (RNNs)** and **Transformer architectures** (including use of GPT-2). Preprocessed parallel corpora (English↔ Spanish) with tokenization and cleaned data pipelines. Built models from scratch and also fine-tuned pretrained transformer models to evaluate translation quality using metrics such as BLEU. Gained experience in neural network design, sequence-to-sequence modeling, attention mechanisms, and model evaluation. Enhanced skills in Python programming, deep learning frameworks (PyTorch or TensorFlow), and handling of large text datasets.

PID control system

Designed and implemented a **PID (Proportional–Integral–Derivative) control system** as part of a **Modern Control Systems project**, focusing on achieving system stability, fast response, and minimal steady-state error. Modeled and analyzed system dynamics using **MATLAB** performing **time-domain and frequency-domain analysis** to tune controller parameters and evaluate performance. The project demonstrated practical understanding of **control theory, system modeling, and feedback mechanisms**, and strengthened skills in **MATLAB simulation, signal analysis, and real-time system optimization**.

Self driving car

Developed a **Self-Driving Car system** that integrates **computer vision, machine learning, and sensor-based decision-making** to enable autonomous navigation. Implemented modules for **lane detection, traffic sign recognition, and obstacle avoidance** using **Python, OpenCV, and deep learning models** such as **Convolutional Neural Networks (CNNs)**. Designed the control logic to process real-time camera input and generate steering and throttle commands for simulated or physical environments. This project enhanced my expertise in **autonomous systems, image processing, and AI-driven control**, while strengthening my understanding of **robotics, perception, and real-time decision-making**.

KBAI book Chatbot

Developed an intelligent **Knowledge-Based AI (KBAI) Book Chatbot** that leverages the **Google Gemini API** to engage in meaningful, context-aware discussions about the “*Knowledge-Based AI: Cognitive Systems*” course content. Integrated **Retrieval-Augmented Generation (RAG)** to combine **LLM reasoning** with a **custom knowledge base** built from course materials and lecture notes. The system retrieves relevant context, processes it through the Gemini API, and generates accurate, explainable responses. This project strengthened my expertise in **NLP, retrieval systems, and LLM integration**, while enhancing my skills in **Python, API development, and AI-driven conversational system design**.